

✅ Playwright + JavaScript + Web + Mobile + API + AI + MCP + Claude – 2025 Master Syllabus

◆ SECTION 1: Software Testing, Playwright & AI Foundations

1.1 Software Testing Fundamentals

- SDLC, STLC
- Manual vs Automation Testing
- Web, Mobile & API Testing Overview
- Agile & DevOps testing

1.2 Introduction to Playwright

- What is Playwright
- Why Playwright in 2025
- Playwright vs Selenium vs Cypress
- Supported browsers & languages
- Use cases in Web, Mobile & API

1.3 Introduction to AI in Testing

- Traditional vs AI-powered testing
- Generative AI in QA
- AI use cases:
 - Test case generation
 - Self-healing
 - Visual AI
 - Smart reporting

1.4 Introduction to MCP Server & Claude

- What is MCP (Model Context Protocol)
 - Why MCP replaces direct AI integrations
 - Claude vs ChatGPT for testing
 - Enterprise AI automation architecture
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◆ SECTION 2: JavaScript COMPLETE Tutorial for Automation

2.1 JavaScript Basics

- Variables: var, let, const
- Data types & coercion
- Operators
- Conditional statements
- Loops

2.2 Functions & Async Programming

- Function types
- Arrow functions
- Callbacks
- Promises
- Async/Await
- Error handling

2.3 Objects, Arrays & Modern JavaScript

- Objects & properties
- Array methods: map, filter, reduce
- Spread & rest
- Destructuring

- Template literals
- Optional chaining

2.4 DOM Manipulation for Automation

- `querySelector`, `querySelectorAll`
- Handling attributes
- Events & listeners
- Shadow DOM basics

2.5 JavaScript Modules & NPM

- `import/export`
- NPM packages
- Writing reusable test utilities

◆ SECTION 3: Playwright Setup & Basics (From Scratch)

3.1 Playwright Installation & Setup

- Node.js installation
- NPM fundamentals
- Playwright installation
- Project structure
- Configuration (`playwright.config.ts`)

3.2 First Playwright Test

- Writing first test
- Running tests
- Headed vs headless
- Debug mode

3.3 Playwright Test Runner Basics

- describe, test, beforeEach, afterEach
 - Assertions with Expect API
 - Auto waits
 - Retries & timeouts
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◆ SECTION 4: Playwright Core Concepts (WEB AUTOMATION)

- Browser, Context & Page
 - Navigation
 - Locators:
 - CSS
 - XPath
 - Text
 - getByRole
 - getByTestId
 - Input actions
 - Checkbox, radio, dropdown
 - File uploads
 - Keyboard & mouse actions
 - Frames & iframes
 - Alerts & dialogs
 - Window & tab handling
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◆ SECTION 5: Playwright for API Automation

- API testing basics
 - HTTP methods
 - `APIRequestContext`
 - `GET`, `POST`, `PUT`, `DELETE`
 - Headers, tokens, cookies
 - Request chaining
 - Response validation
 - API mocking
 - Data-driven API tests
 - API + UI hybrid automation
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◆ SECTION 6: Playwright for Mobile & Device Testing

- Mobile emulation
 - Viewports
 - User agents
 - Geolocation
 - Permissions
 - Touch actions
 - Responsive testing
 - Mobile-first automation strategy
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◆ SECTION 7: Advanced Playwright Features

- Network interception
 - Request/response mocking
 - Screenshots
 - Video recording
 - Tracing
 - HAR recording
 - Downloads & uploads
 - Permissions
 - Timezone testing
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◆ SECTION 8: Framework Design & Test Architecture

- Playwright Test Runner architecture
 - Page Object Model (POM)
 - Hybrid framework
 - Data-driven framework
 - Environment-based execution
 - Parallel & sharded execution
 - Cross-browser strategy
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◆ SECTION 9: AI Test Case Generation (Claude + ChatGPT)

- Manual → AI Test Cases
 - FRS → Test Cases
 - BDD → Playwright Code
 - Edge & negative cases via AI
 - Prompt engineering for QA
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◆ SECTION 10: MCP Server with Playwright & Claude (Enterprise Core)

- MCP server setup
 - Context & tools
 - Claude integration via MCP
 - Test generator agent
 - Debug agent
 - Log analyzer agent
 - CI failure analysis agent
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◆ SECTION 11: Self-Healing Automation Using AI

- Locator healing via Claude
 - DOM change understanding
 - Smart fallback strategies
 - Failure re-execution logic
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◆ SECTION 12: AI Visual Testing

- Visual baselines
 - AppliTools integration
 - Claude image analysis
 - Layout & UI diff detection
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◆ SECTION 13: Smart Reporting & AI Analytics

- Playwright HTML report
 - Allure reports
 - Claude-based root cause reports
 - Failure clustering
 - Slack / Email AI notifications
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◆ SECTION 14: DevOps, Docker & Cloud Execution

- Jenkins integration
 - GitHub Actions
 - GitLab CI
 - Dockerized Playwright
 - Cloud grids
 - MCP-powered CI AI diagnostics
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◆ SECTION 15: Performance, Accessibility & Security

- Lighthouse performance testing
 - Axe-core accessibility
 - OWASP automation
 - AI anomaly detection
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◆ SECTION 16: Real-World Industry Projects

- ✓ AI-Powered E-Commerce Automation
 - ✓ Mobile + Web Hybrid Automation
 - ✓ API + UI Banking Project
 - ✓ MCP Based AI Test Orchestration
 - ✓ Self-Healing Framework Project
 - ✓ Visual AI Regression Project
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◆ SECTION 17: Interview, Certification & Career Support

- Playwright Interview Questions
 - JavaScript Interview Questions
 - API Automation Interview
 - MCP & Claude Interview
 - Resume & Portfolio Building
 - Mock Interviews
 - Industry Certification
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✓ FULL TOOL STACK

- Playwright
 - JavaScript / TypeScript
 - Node.js, NPM
 - Claude
 - ChatGPT
 - GitHub Copilot
 - MCP Server
 - Applitools
 - Jenkins
 - GitHub Actions
 - Docker
 - Allure
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✓ THIS SYLLABUS IS:

- ✓ Beginner to Architect
 - ✓ Enterprise & Startup Ready
 - ✓ AI + MCP Enabled
 - ✓ Web + Mobile + API Unified
 - ✓ CI/CD + Docker integrated
 - ✓ Visual + Performance + Security included
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✅ What You've Already Covered PERFECTLY (Job-Ready Level)

You already fully cover:

- ✅ Playwright Web Automation (Core + Advanced)
- ✅ Playwright API Automation
- ✅ Mobile Emulation with Playwright
- ✅ JavaScript from Scratch
- ✅ Framework Design (POM, Hybrid, Data-driven)
- ✅ CI/CD (Jenkins, GitHub Actions, GitLab)
- ✅ Docker + Cloud Execution
- ✅ AI Test Generation (Claude + ChatGPT)
- ✅ MCP Server + AI Agents (Very Advanced)
- ✅ Self-Healing Automation
- ✅ Visual Testing (Applitools)
- ✅ Performance, Accessibility & Security
- ✅ Real-world End-to-End Projects
- ✅ Interview + Certification + Career Layer

👉 This already qualifies someone for:

- **Playwright Automation Engineer**
- **SDET (UI + API)**
- **Senior Playwright QA**
- **AI Automation Engineer (Junior-Mid)**

